1. Input: time-series spatial transcriptomics

Observed snapshots at times $\{t_k\}_{k=0}^{K}$

Cell states $\{\mathbf{x}_i^{(k)}\}_i$ and spatial coordinates $\{\mathbf{r}_i^{(k)}\}_i$

2. Preprocessing and joint state construction

Normalize expression, select HVGs, reduce dimension (PCA), and align spatial slices

Joint state $\mathbf{z}_i^{(k)} = (\mathbf{x}_i^{(k)}, \mathbf{r}_i^{(k)})$ in $\mathbf{Y} = \mathbb{R}^{d_x} \times \mathbb{R}^{d_r}$

3. Dynamic interaction graph

For each time t, build G_t from spatial proximity and ligand–receptor priors

4. Interaction Velocity

Use an equivariant GNN on G_t to estimate $\mathcal{I}_\theta[\rho_t](\mathbf{z}, t)$

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Interaction-induced transcriptomic and spatial velocity

5. Continuous dynamics

$\mathbf{v}_\theta(\mathbf{z}, t)$ Prob.-flow drift

$\mathbf{s}_\theta(\mathbf{z}, t)$ Score $(\nabla_{\mathbf{z}} \log \rho_t)$

$g_\theta(\mathbf{z}, t)$ Growth rate

6. Reconstruct virtual tissue dynamics

$$\frac{d\mathbf{z}_i}{dt} = \mathbf{v}_\theta(\mathbf{z}, t) + \mathcal{I}_\theta[\rho_t](\mathbf{z}, t)$$

$$\frac{dw_i}{dt} = g_\theta(\mathbf{z}_i, t)\, w_i$$

The coupled ODE system

At time t:

1. From states $\mathbf{z}_i(t) \rightarrow$ build $\mathcal{G}_t$
2. Compute $\mathcal{I}_\theta[\rho_t](\mathbf{z}_i(t), t)$; $\mathbf{v}_\theta(\mathbf{z}_i(t), t)$; $g_\theta(\mathbf{z}_i(t), t)$
3. Update states $\mathbf{z}_i$ and weights w_i (left)
4. Advance to $t + \Delta t$

7. Learning dynamical components

1. Simulate continuous trajectories of states, positions, and population
2. Match generated and observed snapshots, compute ➡ $\mathcal{L}_{\text{OT}}$ Distribution matching (OT in joint space); $\mathcal{L}_{\text{mass}}$ Population matching; $\mathcal{L}_{\text{energy}}$ Variational regularization
3. Update $\mathcal{I}_\theta[\rho_t](\mathbf{z}, t)$; $\mathbf{v}_\theta(\mathbf{z}, t)$; $\mathbf{s}_\theta(\mathbf{z}, t)$; $g_\theta(\mathbf{z}, t)$

Outputs

- Continuous cell-state and spatial trajectories
- Population size and growth patterns
- Lineage trajectories and fate inference
- Spatiotemporal ligand–receptor programs
- In silico perturbation and downstream analysis